# RAMAKRISHNA MISSION VIDYAMANDIRA

**(Residential Autonomous College affiliated to University of Calcutta)**

**M.A./M.Sc. FIRST SEMESTER EXAMINATION, JANUARY 2023**

**FIRST YEAR [BATCH 2022-24]**

**COMPUTER SCIENCE**

**Paper : CC1**

Date : 01/02/2023

Time : 11.00 am – 1.00 pm

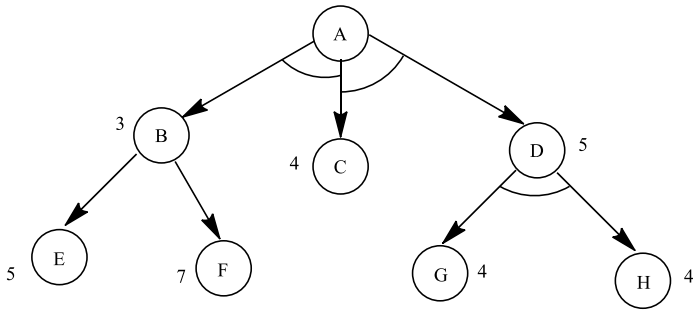
Full Marks : 50

Answer **any five** questions: [5×10]

1. a) Explain how data, information, knowledge and wisdom is related to each other.
   b) What are the approaches of modelling an Artificial Intelligence system?
   c) Comment critically on whether or not you think the Turing test is appropriate for deciding whether or not nonhuman machine can 'think'. Prepare at least one alternative. [3+4+3]

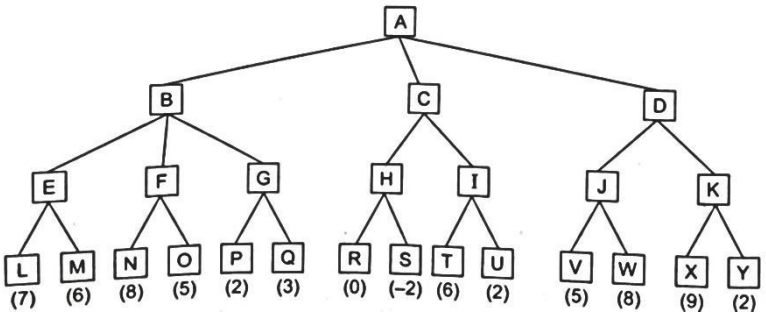
2. a) The goal is to navigate a robot out of a maze. The robot starts in the centre of the maze facing north. You can turn the robot to face north, each south, or west. You can direct the robot to move forward a certain distance, although it will stop before hitting a wall. Formulate this problem using the notion of State Space search.
   b) Give a comparative study on different search techniques with examples. [6+4]

3. a) Compare and contrast between at least two Intelligent agents.
   b) Perform AO* search on the following tree assuming unit weight for each edge:




   c) Is any Heuristic search technique is guaranteed to find the optimal path to solution? Explain your answer. [4+4+2]

4. a) Consider the following game tree in which static scores are all from the first player's point of view:




   Suppose the first player is the maximizing player. Which move should be chosen?
   b) Trace the variables, domains and constraints for the following cryptarithmatic problem:

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  CROSS
 +ROADS
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 DANGER
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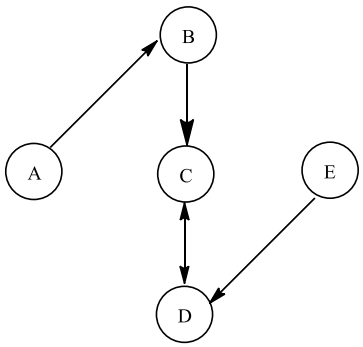
   c) Write down any two real world problems as CSP and explain why they can be classified as CSPs.[4+4+2]

5. a) Convert the following sentences into clausal form:
   i) $A \Leftrightarrow (B \vee E)$.
   ii) $C \wedge F \Rightarrow \neg B$.
   b) What are the advantages of FOPL over propositional logic? Explain with suitable example.
   c) What are the steps to convert an Arbitrary WFF to conjunction of clauses? [(2+2)+3+3]

6. a) Translate the following sentences into first order logic. Define all predicates, functions, and constants you use:
   i) Every person who buys a policy is smart.
   ii) Only one student took Statistics in 2nd Semester.
   b) Write short note on Unification and Substitution. [(3+3)+4]

7. a) Assume the following facts:
   - Ramesh likes only easy courses.
   - Science courses are hard.
   - All the courses in the Sports department are easy.
   - SP201 is a Sports course.

   Use resolution to answer the question, "What course would Ramesh like?"
   b) Represent the following graph in PROLOG. Also write clause to find path between any two vertices:




[5+(3+2)]

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